INSTRUCTOR'S DISCUSSION GUIDE

Topic 13: Network Testing and Troubleshooting

Inspect evidence, run state-derived diagnostic commands, and repair nine common network faults.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Total Discussion Units: 9 Slides

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Systematic Troubleshooting Process

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Troubleshooting starts by defining the symptom and collecting evidence. A justified change is followed by repeating the original failing test and documenting the result.

- Start with power, cable, port, and wireless association. Compare with a known-good link when authorized.
- Check address, mask, duplicate conflicts, Dynamic Host Configuration Protocol (DHCP) service, and gateway. Test local and remote reachability separately.
- If numeric connectivity works but names fail, investigate Domain Name System (DNS). If the host is reachable but a share is denied, inspect identity and permissions.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Start with power, cable, port, and wireless association.
- Compare with a known-good link when authorized.
- Check address, mask, duplicate conflicts, DHCP service, and gateway.
- Test local and remote reachability separately.
- If numeric connectivity works but names fail, investigate DNS.
- If the host is reachable but a share is denied, inspect identity and permissions.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of systematic troubleshooting process.

Key Takeaway: Troubleshooting starts by defining the symptom and collecting evidence. A justified change is followed by repeating the original failing test and documenting the result.

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Six Windows diagnostic commands

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Diagnostic commands inspect configuration, reachability, name resolution, or neighbor information. Each answers a limited question rather than proving that the whole network works.

- ipconfig: shows basic interface configuration; it does not prove reachability.
- ipconfig /all: shows detail such as Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP) server, and leases; it does not prove reachability.
- ping: tests Internet Control Message Protocol (ICMP) Echo behavior and round-trip time; a timeout can also mean filtering or packet loss.
- tracert: provides TTL-based hop evidence; filtered probes can produce gaps.
- nslookup: queries Domain Name System (DNS); success does not prove the destination service works.
- arp -a: lists cached Internet Protocol version 4 (IPv4) neighbor mappings, not every device on the Local Area Network (LAN).

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- ipconfig: shows basic interface configuration; it does not prove reachability.
- ipconfig /all: shows detail such as DNS, DHCP server, and leases; it does not prove reachability.
- ping: tests ICMP Echo behavior and round-trip time; a timeout can also mean filtering or packet loss.
- tracert: provides TTL-based hop evidence; filtered probes can produce gaps.
- nslookup: queries DNS; success does not prove the destination service works.
- arp -a: lists cached IPv4 neighbor mappings, not every device on the LAN.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of six windows diagnostic commands.

Key Takeaway: Diagnostic commands inspect configuration, reachability, name resolution, or neighbor information. Each answers a limited question rather than proving that the whole network works.

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Nine modeled network faults

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Similar symptoms can come from different physical, addressing, service, or permission faults. Distinguishing the cause requires evidence tied to the failed operation.

- Ethernet cable disconnected.
- Incorrect static Internet Protocol (IP) address.
- Incorrect subnet mask.
- Duplicate Internet Protocol (IP) address.
- Dynamic Host Configuration Protocol (DHCP) server unavailable.
- Incorrect default gateway.
- Domain Name System (DNS) server problem.
- File-sharing permission denied.
- Wireless client disconnected.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Ethernet cable disconnected.
- Incorrect static IP address.
- Incorrect subnet mask.
- Duplicate IP address.
- DHCP server unavailable.
- Incorrect default gateway.
- DNS server problem.
- File-sharing permission denied.
- Wireless client disconnected.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of nine modeled network faults.

Key Takeaway: Similar symptoms can come from different physical, addressing, service, or permission faults. Distinguishing the cause requires evidence tied to the failed operation.

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Example: numeric access works but a name fails

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A successful numeric service test and a failed name query point to different functional results. Name-resolution settings and reachability deserve investigation before replacing an already working link.

- If the same service works using its numeric address while name resolution times out, investigate resolver settings and name-service reachability.
- Do not replace a working cable based on this symptom alone. Correct the identified resolver fault, then repeat the original name-based request.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Inspect evidence, run state-derived diagnostic commands, and repair nine common network faults.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- If the same service works using its numeric address while name resolution times out, investigate resolver settings and name-service reachability.
- Do not replace a working cable based on this symptom alone.
- Correct the identified resolver fault, then repeat the original name-based request.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of example: numeric access works but a name fails.

Key Takeaway: A successful numeric service test and a failed name query point to different functional results. Name-resolution settings and reachability deserve investigation before replacing an already working link.

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Understanding Traceroute Results

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Traceroute uses limited-lifetime probes to obtain evidence from successive hops. A missing reply can reflect filtering, while an explicit unreachable response provides different evidence about the path.

Interactive Demonstration: `ttlTraceDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

> *Inspect evidence, run state-derived diagnostic commands, and repair nine common network faults.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Inspect modeled ICMP TTL-expiry evidence, hidden replies, and an explicit broken-link report.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of understanding traceroute results.

Key Takeaway: Traceroute uses limited-lifetime probes to obtain evidence from successive hops. A missing reply can reflect filtering, while an explicit unreachable response provides different evidence about the path.

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Verifying a Network Repair

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A troubleshooting hypothesis should explain the available evidence and predict a useful test. A repair is confirmed by repeating the original failed operation, not by assuming that a changed setting solved it.

- 169.254.x.x suggests link-local configuration; check Dynamic Host Configuration Protocol (DHCP) reachability rather than guessing a public address.
- A wrong gateway can break remote access while local peers still work. A Domain Name System (DNS) fault can leave numeric connectivity intact.
- Document the original state, corrective action, and successful retest. Escalate unfamiliar equipment faults instead of making uncontrolled changes.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 169.254.x.x suggests link-local configuration; check DHCP reachability rather than guessing a public address.
- A wrong gateway can break remote access while local peers still work.
- A DNS fault can leave numeric connectivity intact.
- Document the original state, corrective action, and successful retest.
- Escalate unfamiliar equipment faults instead of making uncontrolled changes.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of verifying a network repair.

Key Takeaway: A troubleshooting hypothesis should explain the available evidence and predict a useful test. A repair is confirmed by repeating the original failed operation, not by assuming that a changed setting solved it.

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Troubleshooting Workflow: Symptom to Fix

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

A complete investigation connects the reported symptom to observations, a likely cause, and a targeted correction. The final retest checks whether the user's original operation now succeeds.

- Define the symptom and scope, then inspect power, physical link, association, and the known healthy configuration.
- ipconfig /all exposes address, mask, gateway, Domain Name System (DNS), and Dynamic Host Configuration Protocol (DHCP) details without proving reachability.
- Compare a local numeric test, a gateway test, a remote numeric test, and a Domain Name System (DNS) query to narrow the affected function.
- Use tracert for hop evidence and arp -a for cached Internet Protocol version 4 (IPv4) neighbors, recognizing filtering and cache limitations.
- Change one justified setting, repeat the original failing test, and document the result. Keep security controls active and escalate unfamiliar equipment faults.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Define the symptom and scope, then inspect power, physical link, association, and the known healthy configuration.
- ipconfig /all exposes address, mask, gateway, DNS, and DHCP details without proving reachability.
- Compare a local numeric test, a gateway test, a remote numeric test, and a DNS query to narrow the affected function.
- Use tracert for hop evidence and arp -a for cached IPv4 neighbors, recognizing filtering and cache limitations.
- Change one justified setting, repeat the original failing test, and document the result.
- Keep security controls active and escalate unfamiliar equipment faults.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of troubleshooting workflow: symptom to fix.

Key Takeaway: A complete investigation connects the reported symptom to observations, a likely cause, and a targeted correction. The final retest checks whether the user's original operation now succeeds.

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Diagnosing Network Faults Workbench

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Different command results expose different parts of the network state. Comparing them supports a targeted repair rather than unrelated configuration changes.

Interactive Demonstration: `troubleshoot` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Modeled network • nine deterministic fault states.
- Inspect commands, select a cause, apply a targeted repair, and retest.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of diagnosing network faults workbench.

Key Takeaway: Different command results expose different parts of the network state. Comparing them supports a targeted repair rather than unrelated configuration changes.

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Local LAN vs Internet vs DNS

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Reaching a local gateway does not prove that remote routes, name resolution, or a website work. Separate tests identify which stage succeeds and which still needs investigation.

- Check numeric external reachability and Domain Name System (DNS) responses to separate routing from name resolution.
- Do not assume the cable is faulty when the gateway is responding.
- Even working Domain Name System (DNS) and ping do not prove a web service or permission is functioning.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Check numeric external reachability and DNS responses to separate routing from name resolution.
- Do not assume the cable is faulty when the gateway is responding.
- Even working DNS and ping do not prove a web service or permission is functioning.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of local lan vs internet vs dns.

Key Takeaway: Reaching a local gateway does not prove that remote routes, name resolution, or a website work. Separate tests identify which stage succeeds and which still needs investigation.